

# Nicholas A. Henry

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## EDUCATION

**Northeastern University** Boston, MA  
*Bachelor of Science in Mechanical Engineering and Computer Science* May 2029  
**Coursework:** Calculus III, Discrete Structures, Program Design and Implementation **GPA: 3.89**  
**Activities:** Deans List, Northeastern Electric Racing FSAE, Students for the Exploration and Discovery of Space (NEST)  
*Studied A Semester Abroad At Queen's University Belfast in Northern Ireland*

## SKILLS

**Technical Skills:** Python, Flask, TensorFlow, Dash Plotly, C#, ROS, Java, JavaScript, HTML/CSS, NumPy, Linux, Docker, REST APIs, Git, Supabase, SQL, GitHub Actions, WordPress, React

**Engineering Skills:** Project Management, Fusion360, SolidWorks, AutoCAD, MATLAB, Problem Solving, Cross-Functional Collaboration, 3D Printing, Laser Cutting, CNC Milling, MIG and TIG Welding

**Certifications & Training:** IBM Artificial Intelligence Fundamentals, Microsoft C# Certification

## EXPERIENCE

**Northeastern Electric Racing - Python, Dash Plotly, React, Docker, Git** Boston, MA  
*Lead Software Engineer for Simulation Team* Jan 2026 – Present

- Lead on the simulation team developing a multi-body vehicle simulation tool for NER engineers to test and examine how changes to geometry and other variables go on to affect the speed and times of the car on custom created tracks
- Utilise Python as the backend to do all the physics with Dash Plotly and React for the frontend of the site to easily see all graphs and metrics worked on creating a large data structure of all the vehicle's variables to allow a tree visualization of dependency

**Students for the Exploration and Discovery of Space - Python, ROS 2, Git, Linux, Docker** Boston, MA  
*Software Engineer for Network Exploration Swarm Toolkit Team* Jan 2026 – Present

- Operate in NEST, a project to create a swarm of miniature C.O.B.R.A. (Crater Observing Bio-inspired Rolling Articulator) snake robots to explore and develop multi-agent collaboration in order to complete teamwork based tasks
- Developed the camera sensor ecosystem using a monocular simultaneous mapping program for multi-agent systems

**SEONeeded - Wordpress, SEOMoz, Semrush** Miami, FL  
*Web Optimization Engineer* Oct 2025 – Aug 2026

- Perform client-facing tasks like identifying keywords, website maintenance, and strategies to target in order to drive more traffic for certain services increasing click-on rate and sales for 15 clients across the Greater South Florida region
- Adapted to the new wave of generative engine optimization space by learning what makes LLM models reference client's websites and how certain worded prompts get tailored responses, recommendations, and links to sources

*Search Engine Optimization Intern* Jun 2025 – Sep 2025  
Drove a 54% increase in website traffic through targeted optimization by leveraging WordPress and keyword research

## PROJECTS

**Portfolio Website - HTML/CSS, JavaScript, SQL, Supabase, GitHub Actions, Git** Present

- Built a site with HTML/CSS and Javascript hosted on Github Pages with a custom domain to show myself and projects
- Applied SQL databases with Supabase in order to have a global comment section where visitors can write something
- Use GitHub Actions in order to autoupdate the resume on the site to match this document as it changes in Google Docs

**Text Transcriber - Python, TensorFlow, Google APIs, Git** Jul 2025

- Designed and trained an AI model using TensorFlow on 100,000+ character images to recognize and convert handwriting into text and to then convert into audio with Google TTS as part of a 3-person team
- Tested with a group of visibly impaired people to see and hear writing and signage enabling them to be less dependant on caretakers in everyday life, with successful results in the limited cohort of 88% understanding the responses

**Sheep Finder - TypeScript, JavaScript, React, Git** May 2026 - Present

- Developed a React web app based on my experience abroad in Northern Ireland designed to assist farmers and create an ecosystem where people can report sightings of sheep making it easier for farmers to pick them up
- Spoke with farmers from Northern Ireland to gain an insight into what is the main issue and the best ways to solve them and built an app around that feedback like how sheep are designated across farms
- Deployed a site demo version which will be turned into an app but have already had 2 towns in Ireland pick it up and had at least 20+ documented uses and even heard back from a farm owner about a success story